

The Great American Star "Birthday Problem"

How quickly does the chance of a matching path grow as more students participate?

Name: _____

Date: _____



The big question

The Great American Star has

359,277,250

possible mathematical paths

If students independently choose paths at random, how likely is it that **two students** choose the same path?

Connection: This activity borrows the idea behind the classic Birthday Problem and applies it to Great American Star paths.

Predict before calculating

Which do you think is more likely for a group of 20 students: at least two sharing a birthday, or at least two choosing the same Great American Star path?

☐ Same birthday

☐ Same star path

☐ About the same

Why do you think so?

Our mathematical model: every path is equally likely, and each student's choice is independent of the others. The calculation describes this model; it does not guarantee how students will actually choose in the creator.

From 2 Students to 20 Students

Two students: one pair

The first student can choose any path. For the second student to match, exactly **one** path works.

1 in 359,277,250

chance that the second path matches the first

Approximately 0.000000278%

Twenty students: how many pairs?

Each of the 20 students can be paired with 19 others. But that counts every pair twice.

$$20 \times 19 \div 2 = 190$$

190 pairs

190 opportunities for a match

Why divide by 2?

Explain why Alex-Sam and Sam-Alex should count as the same pair.

With 20 students

0.0000529%

chance of at least one matching pair

1 in 1.89 million

approximately

Even though 20 students create **190 pairs**, an exact Great American Star path match remains extraordinarily unlikely because the pool contains 359,277,250 possibilities.

What changed when we went from 2 students to 20 students?

Same 190 Pairs. Very Different Chances.

Now compare Great American Star paths with birthdays. Assume 365 equally likely birthdays, ignore leap day, and keep the same 20 students.

20 students	Birthdays	Great American Star paths
Student-pairs	190	190
Possible choices	365	359,277,250
Chance of at least one match	41.1%	0.0000529%

SAME 190 PAIRS.
Very different size of the possibility pool.

1

Why is a birthday match so much more likely?

2

Why does the star-path probability remain tiny?

3

Which two quantities control the chance of a match?

4

What do you predict for 30, 50, or 100 students?

The key idea

Two quantities matter:

1. How many pairs can be formed.

2. How large the pool of possible choices is.

More pairs increase the chance of a match; a much larger possibility pool makes each exact match much less likely.

Connect it back to the Star

When you create your own Great American Star, your chosen path is one choice among **359,277,250** mathematical possibilities.